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Late Cretaceous paleomagnetism of the East Ranges island arc complex, Kamchatka: Implications for terrane movements and kinematics of the northwest Pacific

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Abstract. A Campanian-lower Paleocene island arc complex was sampled for paleomagnetic studies at 12 sites in the East Ranges tectonic zone of Kamchatka. After thermal demagnetization, a reversed polarity characteristic remanent magnetization (ChRM) was isolated from most volcanoclastic and basaltic units as well as from lava debris from intraformational conglomerates. The fold and conglomerate tests are positive, and the ChRM in the studied rocks is likely primary. The formation-mean inclination of 66.3° ± 3.7° corresponds to a paleolatitude of 48.7° ± 5.0°N which is about 20° lower than the Late Cretaceous North American reference values. Because northward displacement of the studied terrane is indicated by the paleomagnetic data, we examine several models of intraoceanic transport with the Pacific and/or Kula plates and coastwise transport after terrane accretion, far to the south of the present-day position of Kamchatka. Our preferred interpretation is that the studied island arc complex accumulated at about 83-79 Ma; the island arc, to which the studied terrane had originally belonged, was active between this time and 65-60 Ma. According to geological data, the docking time nearly coincided with cessation of volcanic activity, and northward movement of the island arc took place simultaneously with the volcanic activity. The absolute motion of a subduction zone should have the same direction as the overriding plate; therefore, the subduction zone related to the East Ranges island arc is inferred to have moved northward with the Kula plate or with the Kula and Pacific plates, successively, consuming either the oceanic periphery of a continental plate or some unknown minor oceanic plate. This process went on until 65-55 Ma when the island arc and related subduction zone approached the continental margin and became extinct. The proposed models also place additional constraints on kinematics of the Kula-Pacific transform plate boundary.

Introduction

Since the early 1980s, the tectonic evolution of northeast Asia and the entire circum-Pacific orogenic belt has been interpreted according to the concept of tectonostratigraphic terranes [*Marlow and Cooper*, 1983; *Debiche et al.*, 1987; *Stavsky et al.*, 1990; *Sokolov*, 1992]. The term "tectonostratigraphic terrane" presumes that a tectonostratigraphic unit, which is different from adjacent units, has been transported over a considerable distance and juxtaposed alongside other units that had different origins. While identification of a terrane and its boundaries relies mainly on geological data, the original position of a terrane and its subsequent kinematics are derived mostly from paleomagnetic data.

Such joint geological and paleomagnetic studies have played an important role in deciphering the tectonic evolution of western North America [e.g., *Bruns*, 1983; *Geist et al.*, 1988; *Harbert et al.*, 1990; *Bol et al.*, 1992; *Bol*, 1993]. In contrast, few studies have been conducted in northeast Asia, particularly in Kamchatka. Interpretations on the number of independent terranes, not to mention their original positions and subsequent horizontal movements, are controversial [*Watson and Fujita*, 1985; *Tsukanov*, 1991; *Geist et al.*, 1994; *Shapiro*, 1995]. Paleomagnetic data can resolve many of these controversies by providing quantitative estimates of horizontal movements to test competing kinematic scenarios. Studies have been made in easternmost Kamchatka (East Peninsulas zone) [*Bazhenov et al.*, 1992; *Pechersky et al.*, 1997] and further to the north on Karaginsky Island and Koryakia [*Kovalenko*, 1990, 1992], with most results being based on fully demagnetized data with confirmations from field tests. The available data from the main part of Kamchatka are of lower quality [*Gurevich and Surkis*, 1990, 1994], and Late Cretaceous paleolatitudes within the same structure vary from nearly equatorial to about 43°N, making their tectonic significance unclear.

We present new paleomagnetic results from Upper Cretaceous volcanoclastic rocks (Khapitsa Formation) in east Kamchatka which may help to resolve some of the existing controversies.

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Tectonic Zones of Kamchatka

Kamchatka is divided into a number of NNE-SSW trending tectonic zones (Figure 1). The West Kamchatka zone comprises

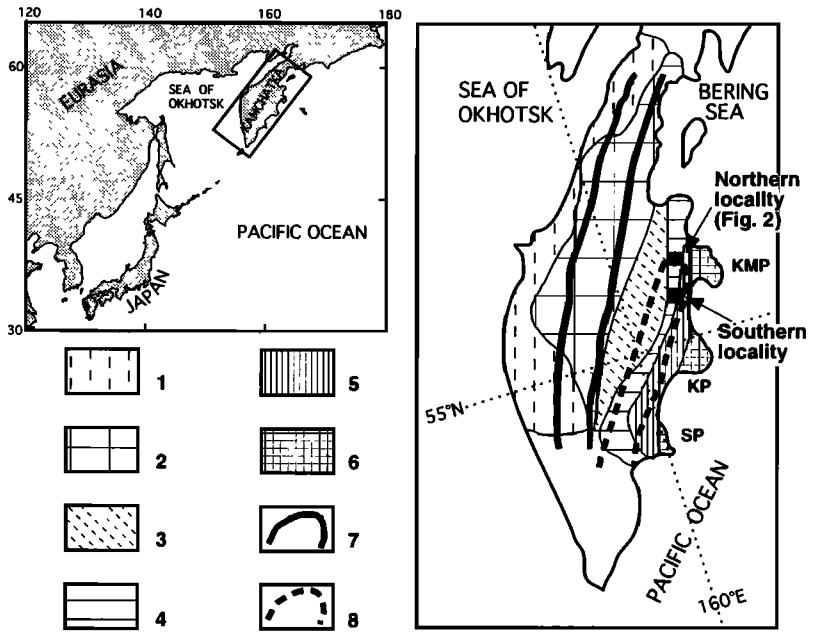


Figure 1. Location map and tectonic map of Kamchatka. Legend: 1, West Kamchatka zone; 2, Median Range zone; 3, Central Kamchatka late Cenozoic basin; 4, western band of the East Ranges zone; 5, eastern band of the East Ranges zone (Vetlov subzone); 6, East Peninsulas zone; 7, Miocene-Quaternary arc volcanic province; 8, Pliocene-Quaternary arc volcanic province. KMP, KP, and SP stand for the Kamchatsky Mys, Kronotsky and Shipunsky Peninsulas, respectively.

Campanian-Maastrichtian terrigenous rocks that accumulated on the continental slope [*Shapiro*, 1976]. Paleogene to Neogene rocks in this zone unconformably overlie Upper Cretaceous sediments and are represented by coal-bearing molasses, shelf sediments, and minor amounts of subaerial volcanics. To the east, the Median Range and East Ranges zones consist of Upper Cretaceous-lower Paleocene island arc complexes. The Median Range island arc complex is unconformably overlain by Eocene to Oligocene marine sediments and Miocene to Quaternary volcanics. This complex is thrust over the West zone along the Lesnov Thrust, which was active in the interval from 65-45 Ma. The boundary between the Median Range and East Ranges island arc complexes is hidden under sedimentary cover of the late Cenozoic Central Kamchatka basin. The easternmost (East Peninsulas) zone consists of Upper Cretaceous and Paleocene-Eocene volcanoclastic rocks of island arc affinity [*Zinkevich and Tsukanov*, 1992] on the Kamchatsky Mys, Kronotsky and Shipunsky Peninsulas, along the eastern coast of Kamchatka (Figure 1). The latter zone is separated from the Kamchatka mainland by a basin that is filled with Oligocene to Miocene marine sediments. The Kamchatsky Mys Peninsula is also regarded as the westernmost part of the Aleutian chain [*Markov et al.*, 1972; *Geist and Scholl*, 1994] or as a separate tectonostratigraphic terrane [*Watson and Fujita*, 1985]. Eocene deformation (ca. 40 Ma) affected almost all of Kamchatka, with only minor deformation in the Miocene and Pliocene. In contrast, the main folding and thrusting events are of Miocene age in the basin that separates the East Ranges and East Peninsulas zones.

Late Cretaceous island arc complexes are known from the Median Range and East Ranges zones. A Paleogene-Eocene island arc complex is recognized only in the peri-Pacific East Peninsulas zone. Oligocene-Miocene island arc volcanism occurred inland, partly overlapping the West and Median Ranges zones. Finally, in the Pliocene-Quaternary, island arc volcanism shifted eastward to its present-day position in east Kamchatka (Figure 1).

The Kumroch Range makes up the northern part of the East Ranges zone, ERZ (Figure 1) which comprises strongly deformed Campanian-Maastrichtian to Paleocene-lower Eocene rocks that are unconformably overlain by weakly deformed middle Eocene to Miocene marine sediments and almost flat-lying Pliocene to Quaternary volcanics and continental deposits. In the west the ERZ is bounded by subvertical faults that separate it from the central Kamchatka depression, which is filled by Pliocene-Quaternary volcanics and continental deposits. Upper Cretaceous-lower Paleogene rocks of the ERZ are deformed into tight isoclinal folds and thrusts with southeastern vergence. The steeply dipping Vetlov Thrust divides this complex into two subparallel bands. The eastern band (Vetlov subzone) comprises Paleocene-lower Eocene siliceous-tuffaceous clay stones with basalt lenses of oceanic and marginal basin affinity (Vetlov Formation). The western band comprises thick Upper Cretaceous volcanics of island arc affinity (Khapitsa Formation and its counterparts) which vary considerably both along and across strike as well as between thrust sheets [*Shapiro et al.*, 1997]. The base of the sequence is never exposed; estimates of the age of the volcanic series are based on rare radiolaria in siliceous interbeds which indicate a late Campanian or Campanian-Maastrichtian, sometimes to Maastrichtian-Danian, age [*Tsukanov*, 1991]. In the south of the Kumroch Range, the Khapitsa Formation is overlain, conformably and with gradual transition, by upper Paleocene terrigenous flysch. Folding and thrusting took place in the middle Eocene.

Geochemically, the Upper Cretaceous-lower Paleocene volcanic sequence of the ERZ is similar to volcanics of intra-oceanic island arcs like the Tonga arc [*Magakyan et al.*, 1993].

This sequence resembles contemporaneous volcanics of the Median Range zone of Kamchatka and the Olutor zone of Koryakia; these three areas are sometimes regarded as a single, wide island arc [*Zinkevich and Tsukanov*, 1992; *Shapiro*, 1995].

Sampling

We studied the Late Cretaceous Khapitsa Formation in the northern and southern parts of the Kumroch Range (Figure 1). Several thrust sheets with southeast vergence are recognized in the north (Figure 2). Some thrust sheets contain evidence of penetrative deformation like cleavage and boudinage, and are not suitable for sampling. Using lithological characteristics, the thickest thrust sheet can be divided into three members with gradual transitions between them. The lower member, which is about 1200 m thick, comprises distinctly interbedded greenish gray tuffaceous rocks. The beds generally dip to the northwest, with variable azimuth and dip. These rocks are fractured but have no cleavage, and no systematic grain reorientation was observed. Samples were taken from sites 1, 2, 5, and 11 in the upper part of this member (Figure 2, Table 1); the exact stratigraphic relationships of these sites, however, are unknown.

The middle member, which is 1500 to 1800 m thick, consists mainly of conglomerates, agglomerated basalts, and andesitic-basaltic tuffs with subordinate interbeds of finer-grained tuffs that are cut by thin sills. These sills apparently formed almost contemporaneously with host rocks, because petrologically similar volcanics are abundant in conglomerates both below and above these sills. Samples were taken from the three sills and their baked contacts (sites 6, 7, and 10), as well as from the fine-grained tuffs (sites 8 and 9). The thin stratigraphic intervals studied at each site (Table 1) are separated by much thicker layers of coarse-grained rocks. Volcanic debris, which is petrograpically similar to that of the sills, was sampled from a thick conglomerate layer (site 4).

The upper member, which is about 1000 m thick, consists of irregularly interbedded tuffaceous beds of varied grain size. The uppermost fine-grained part of the section was either not exposed in the studied area or was metamorphozed by Pliocene intrusions. As the result, only the lower portion of the upper member was sampled (site 12).

Site 3 was sampled from another, structurally lower, thrust sheet (Figure 2) where the division into three members cannot be made, and the rocks differ from the upper thrust sheet in color and are more zeolitized.

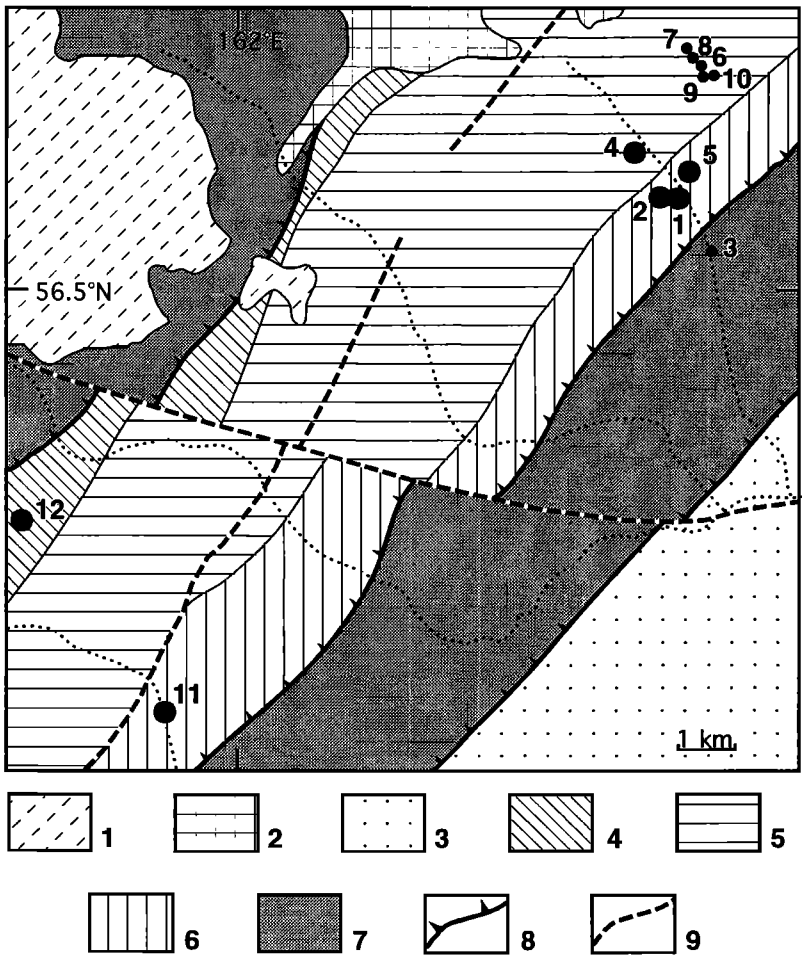


Figure 2. Schematic geological map of the northern locality in the Kumroch Range. Legend: 1, Neogene volcanics; 2, Neogene intrusions; 3, Vetlov Formaton; 4-6, Studied thrust sheet (4, Lower member; 5, Middle member; 6, Upper member); 7, Lower thrust sheet; 8, Thrust faults; 9, Other faults. Sampling sites are shown as solid dots and are numbered as in the text. Large (small) dots denote sites accepted (rejected) for interpretation.

Table 1. Paleomagnetic Results From the North Kumroch Range, Upper Thrust Sheet

				In Situ				Tilt Corrected			
S	N	Th, m	B	D, deg	I, deg	k	a95, deg	D, deg	I, deg	k	a95, deg
				Lower Member							
1	41/27	20	318/54	308.2	-59.7	18	6.4	150.5	-65.7	23	5.6
2	29/15	40	313/73	308.2	-45.3	15	9.5	146.5	-60.6	16	9.1
5	24/20	100	276/59	253.4	-39.1	14	8.3	156.6	-70.0	14	8 2
11	18/9	100	285/80	267.7	-28.8	21	10.1	143.6	-65.8	36	7.8
Sample mean	112/71			282.6	-49 6	9	5.6	150.0	-65.9	19	3 8
Site mean	4			280.8	-45.8	12	27.7	148.9	-65.6	333	5.0
				Middle Member							
10	9/8	2	345/84	355.6	-53.6	22	10.6	156 6	-41.7	22	10 6
9	16/14	7	326/63	316.3	-52.3	19	8.7	158.6	-63 5	16	9.3
8	8/8	2	347/79	338.5	-10.9	28	9.4	257 1	-81.9	40	7 8
7	15/13	3.5	338/71	347.8	-20.3	15	10.0	77.3	-80.4	16	9 7
6	32/20	5	339/62	318.3	-21.5	42	4.8	263.3	-70.2	46	4.6
Sample mean	80/63			331.2	-31.5	10	5.7	186.3	-78.3	9	5 8
Site mean	5			335.2	-32.5	11	23.7	172.6	-75.7	11	24.7
				Upper Member							
12	25/22	180	322/79	302.2	-22.8	14	8.1	199.5	-68.4	16	7 6
				Overall Mean (Sites 1, 2, 5, 11, and 12 Combined)							
Sample mean	137/93			288.7	-43.5	8	5.2	160.1	-67.8	16	3.7
Site mean	5			285 6	-40.8	12	18.4	157 5	-67.4	73	7.3
				Inclination-Only Statistics							
Samples	137/93				-48.1	4	20.2		-64.3	26	2.9
Sites	5				-40.7	10	18.8		-66.3	182	3 7

Column headings are S, site number (see text and Figure 2); N, number of samples studied/accepted, Th, stratigraphic thickness studied; B, azimuth of dip/dip angle of bedding; D, declination; I, inclination; k, Fisher concentration parameter; a_{95}, radius of 95% confidence circle.

The southern locality (Figure 1; a separate map is not given) is characterized by tight N-S trending folds with subvertical axial planes. The cores of the anticlines and synclines are composed of the Khapitsa Formation and terrigenous flysch, respectively. The main folds are cut by numerous small faults, with conjugate short-wavelength folds. In general, the Upper Cretaceous rocks resemble those of the middle member at the northern locality, but zeolitization is stronger, and fewer coarse-grained rocks are exposed; in particular, the conglomerates are absent. Seventy five samples were taken from fine-grained tuffs at eight sites.

The true stratigraphic thickness of the studied outcrops, which were treated here as sites, varies from several meters to about 200 m (Table 1). One hand block, oriented with a magnetic compass, was taken from each stratigraphic layer, with samples being spaced more or less uniformly across the sampled outcrops. Between one and four (usually three) cubic specimens were cut from each block in the laboratory.

Laboratory Treatment

One to three cubic specimens from each hand sample were thermally demagnetized in 12-16 steps in home-made ovens with internal residual fields of about 10 nT. All measurements were made on a JR-4 spinner magnetometer with a noise level of 0.05 mA/m. The characteristic remanent magnetization (ChRM) was determined without anchoring the final linear segments to the origin of vector component diagrams [*Kirschvink*, 1980]. ChRMs isolated from sister specimens of a hand sample were used to calculate sample means, which, in turn, were used to calculate site means.

Results

Apart from an unstable remanence removed below 300°C to 350°C, no consistent directions were isolated from Upper Cretaceous rocks from the southern locality, and the entire collection was discarded. In contrast, similar rocks from the northern locality yielded much better data; these data, however, are of variable quality.

Lower Member of the Khapitsa Formation (Sites 1, 2, 5, and 11)

After removal of an unstable remanence at 200°C, the natural remanent magnetization (NRM) in most greenish gray tuffaceous rocks is accounted for by a single ChRM above 250°-300°C (Figure 3a). The ChRMs generally display a rectilinear decay to the origin of the vector component diagram, although some are not directed to the origin (Figures 3a and 3b). The directions, on average, remain the same irrespective of whether the linear segments of the ChRM pass through the origin. Complete demagnetization of these rocks, however, was often prevented by acquisition of spurious remanence components above 500°C.

In some samples, another remanence component is present at low to intermediate temperatures (Figure 3b). Although represented by rectilinear segments on demagnetization plots, this remanence probably results from partial overlap of unblocking temperature spectra of a recent overprint and the ChRM. Judging by maximum unblocking temperatures, the ChRM appears to reside dominantly in magnetite. After tilt correction, the ChRM points upward to the southeast except for two samples where this component is of opposite polarity. The

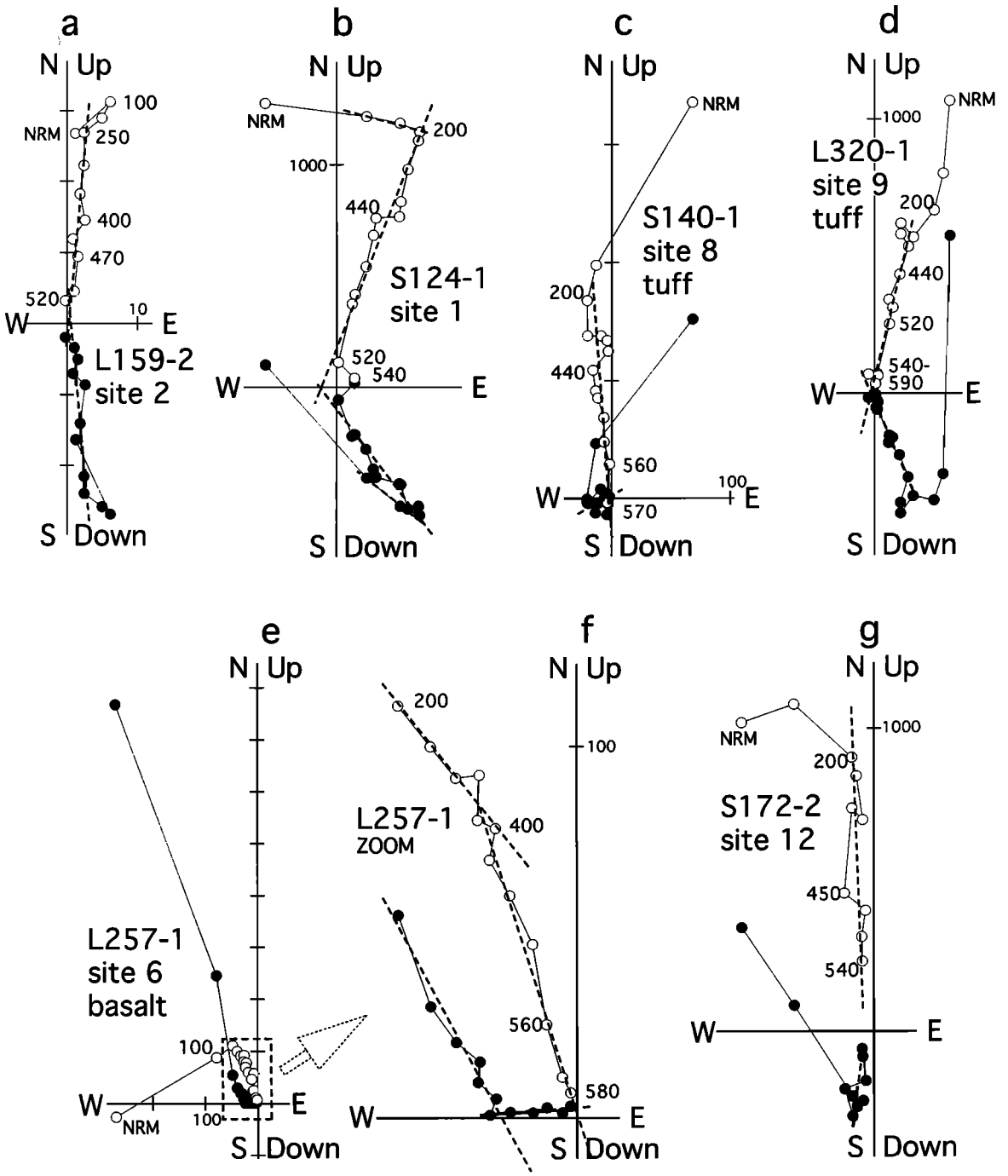


Figure 3. Representative vector component plots for Upper Cretaceous tuffaceous rocks from the (a,b) lower, (c-f) middle, and (g) upper members of the Khapitsa Formation. Solid (open) symbols are projected onto the horizontal (vertical) plane. The plots are in stratigraphic coordinates. Isolated components are shown as dashed lines. Temperatures are in degrees Celsius. Magnetizations are in milliamperes per meter.

ChRM directions are well grouped at each site, and the site means are well defined (Table 1). The dispersion of site means is greatly reduced after tilt correction (Figures 4a and 4b; Table 1), and the fold test [*McFadden and Jones*, 1981] is positive at the 95% confidence level (Table 2).

Middle Member of the Khapitsa Formation (Sites 6-10)

After removal of a viscous overprint, a ChRM was isolated above 300°C from basalts and tuffaceous rocks (Figures 3c and 3d). In other samples, another rectilinear segment can be recognized at low to intermediate (up to 400°C) temperatures, mainly in samples where the NRM intensity decreased markedly at the initial demagnetization levels (Figures 3e and 3f). As for the lower member of the Khapitsa Formation, we attribute this behavior to the partial overlap of unblocking spectra of a recent overprint and the ChRM. We therefore exclude the low- and intermediate-temperature components from consideration.

The ChRM in most samples persists up to 560°-580°C, which implies that magnetite is the main ChRM carrier. ChRM

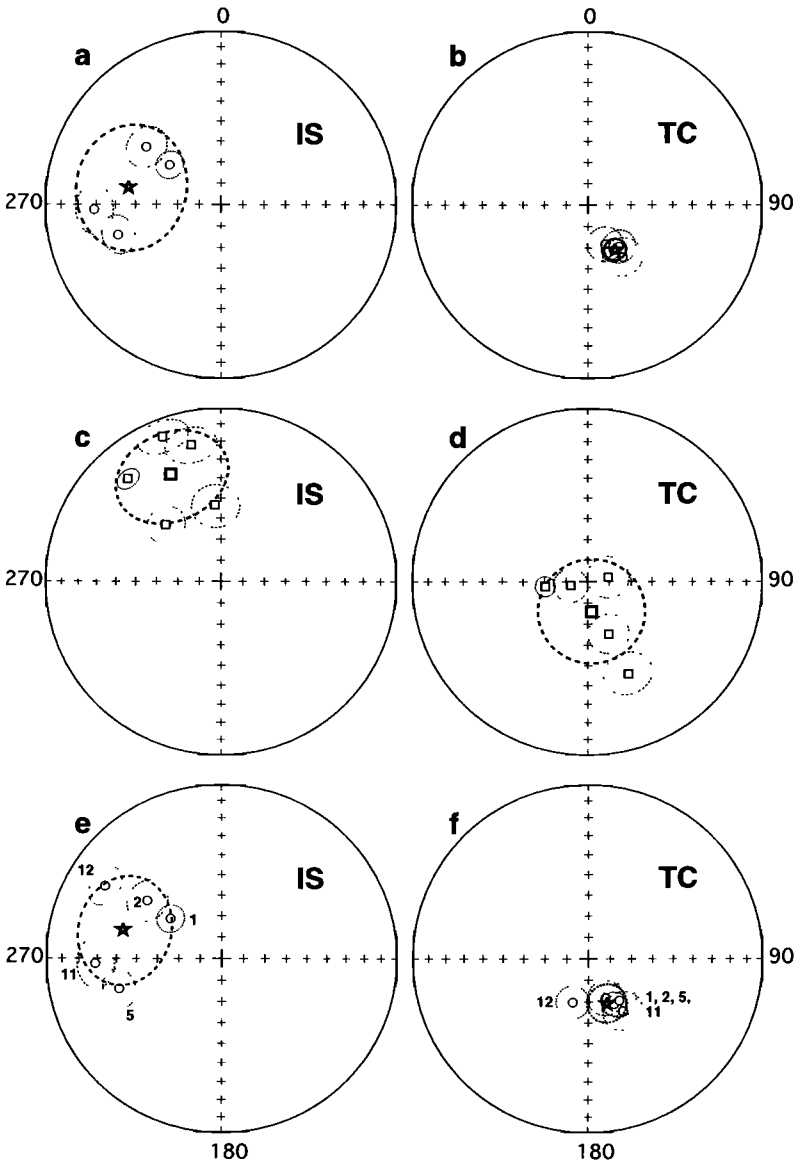


Figure 4. Stereoplots of ChRM site mean directions for the Upper Cretaceous Khapitsa Formation in situ (IS) and after tilt correction (TC): (a, b) lower member, (c, d) middle member, (e, f) lower and upper members combined. For the lower and upper members, open circles and thin dotted ellipses are site means and confidence limits, respectively; stars and thick dotted ellipses are the overall means. For the middle member, small open squares and thin dotted ellipses are site means and confidence limits, respectively; large open squares and thick dotted ellipses are overall means with associated confidence limits. All symbols are projected onto the upper hemisphere.

directions are fairly well grouped for each site, but the site means are dispersed both before and after tilt correction (Figures 4c and 4d; Table 1); no noticeable improvement in data grouping was observed during incremental unfolding. The fold test is therefore inconclusive for this group of sites.

Table 2. Fold Test Results From the North Kumroch Range, Upper Thrust Sheet

	Critical values	Calculated statistics	
		In Situ	Tilt corrected
Lower member	$F_{(6,134)} = 2.1$	20.5	0.9
Overall mean	$F_{(8,176)} = 1.99$	23.0	4.45

The number of degrees of freedom of F statistics is given in brackets

A 2-m-thick tuff layer and a 3-m-thick lower part of a basalt sill, along with the intervening baked contact, was studied at site 6. After demagnetization, the basalt samples from more than 1 m from the contact yielded noisy data and dispersed directions. In contrast, ChRM directions of the tuffs, the baked contact, and the lower 50-70 cm of the sill are well grouped (Table 1). However, this does not constitute a positive baked contact test because paleomagnetic directions are the same for each material.

Upper Member of the Khapitsa Formation (Site 12)

After removal of a low-temperature remanence component, a reversed polarity component was isolated from most samples above 300°C or 350°C (Figure 3g). Complete demagnetization

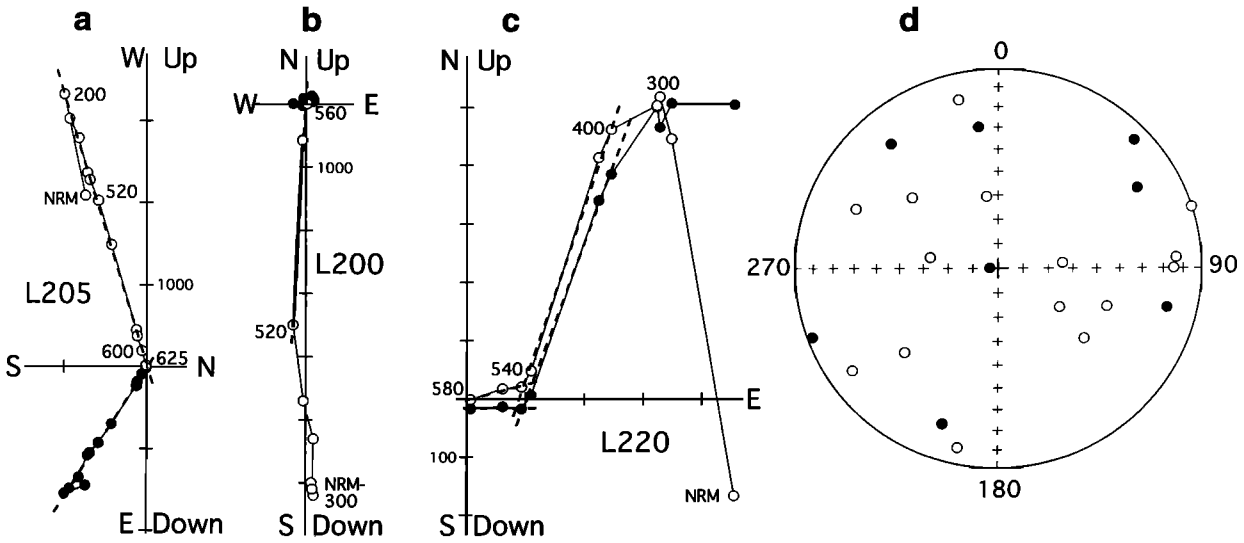


Figure 5. (a-c) Representative vector component plots and (d) stereoplot of ChRM directions from intraformational conglomerate boulders (site 4). For Figures 4a to 4c, conventions are the same as in Figure 3. For Figure 4d, solid (open) symbols are projected onto the lower (upper) hemisphere.

of these rocks, however, was often precluded by acquisition of spurious components at about 500°C; nevertheless, ChRM directions are well grouped, and the site mean is well defined (Table 1). The fold test is inconclusive due to limited variation of bedding attitudes at this site.

Lava Boulders From an Intraformational Conglomerate (Site 4)

The conglomerate test was performed on 24 lava boulders from site 4 (Figure 2). The NRM in the boulders contains a low-temperature overprint with a dominant linear component that decays to the origin of the diagram (Figures 5a and 5b). In a few samples, another component is present at intermediate temperatures (Figure 5c), and the high-temperature component was interpreted as the ChRM. ChRM directions are chaotically distributed (Figure 5d), and the normalized vector-resultant of 0.265 is much less than the critical limit of 0.334 for a random distribution [*Mardia*, 1972]. Site 4, where the lava boulders were sampled, lies within the middle member of the Khapitsa Formation, near sites 1, 2, and 5 in the lower member. On the other hand, the rocks from the lower and upper members show similar demagnetization characteristics. Thus we interpret the positive conglomerate test to be indicative of a primary age for the ChRM which we interpret to be valid for the entire upper thrust sheet.

Khapitsa Formation From the Lower Thrust Sheet (Site 3)

After removal of an unstable remanence component at 200°-300°C, a well-defined ChRM was isolated from most samples from site 3, with almost all orthogonal plots decaying to the origin (Figures 6a to 6e). The ChRM persists up to 560°-570°C, which indicates that the ChRM resides in magnetite.

Site 3 is the only site where the ChRM of many samples has normal polarity in stratigraphic coordinates, but ChRM directions are scattered (Figure 6f). There is a cluster of northeast and downward pointing directions, with several samples having southeast declinations and either positive or negative inclinations. The streaked distribution of directions hints at a possibility of a considerable overprint which remains unresolved despite the apparently good demagnetization data. Whatever the origin of this pattern, we did not include these data in further analyses.

Combined Data

Site means from the lower and upper members of the Khapitsa Formation show considerable improvement in grouping after tilt correction. However, a more rigorous comparison [*McFadden and Jones*, 1981] indicates that the mean of site 12 is statistically different from the other four directions: the calculated F statistic is 4.45, whereas the corresponding critical value is 1.99 (Table 2). Because site 12 is spatially and stratigraphically distant from the other sites, and because the difference is mainly in declination (Figures 4e and 4f), we attribute this divergence to a clockwise rotation or a tilt of the area around site 12 relative to the other studied areas. The positive fold and conglomerate tests indicate that the ChRM in these members of the Khapitsa Formation is probably primary. For calculation of the mean inclination for these members, the inclination-only statistics [*McFadden and Reid*, 1982] was applied to obtain the value that is used in later analyses (Table 1).

Poorer consistency of results from the middle member of the Khapitsa Formation cannot be attributed to local rotations because all of the sites were sampled close to each other (Figure 2), and site means differ in both declination and inclination (Figures 4c and 4d, Table 1). This member consists mainly of conglomerates and agglomerated tuffs with rare, thin, fine-grained tuffs and sills. This, and scarcity of outcrops, enabled us to sample only thin beds at each site (Table 1). Tuffaceous rocks can accumulate rapidly, and deposition of a 2-m-thick layer could occur (geologically) almost instantaneously. Cooling of a few-meter-thick sill and its baked contact to well below the Curie temperature also takes a relatively short time. We hypothesize that the large scatter of site-means from this member is due to poor averaging of secular variation within each site; the observed grouping of these five site mean directions does not contradict the above explanation. If the above explanation is correct, the site means from the middle member

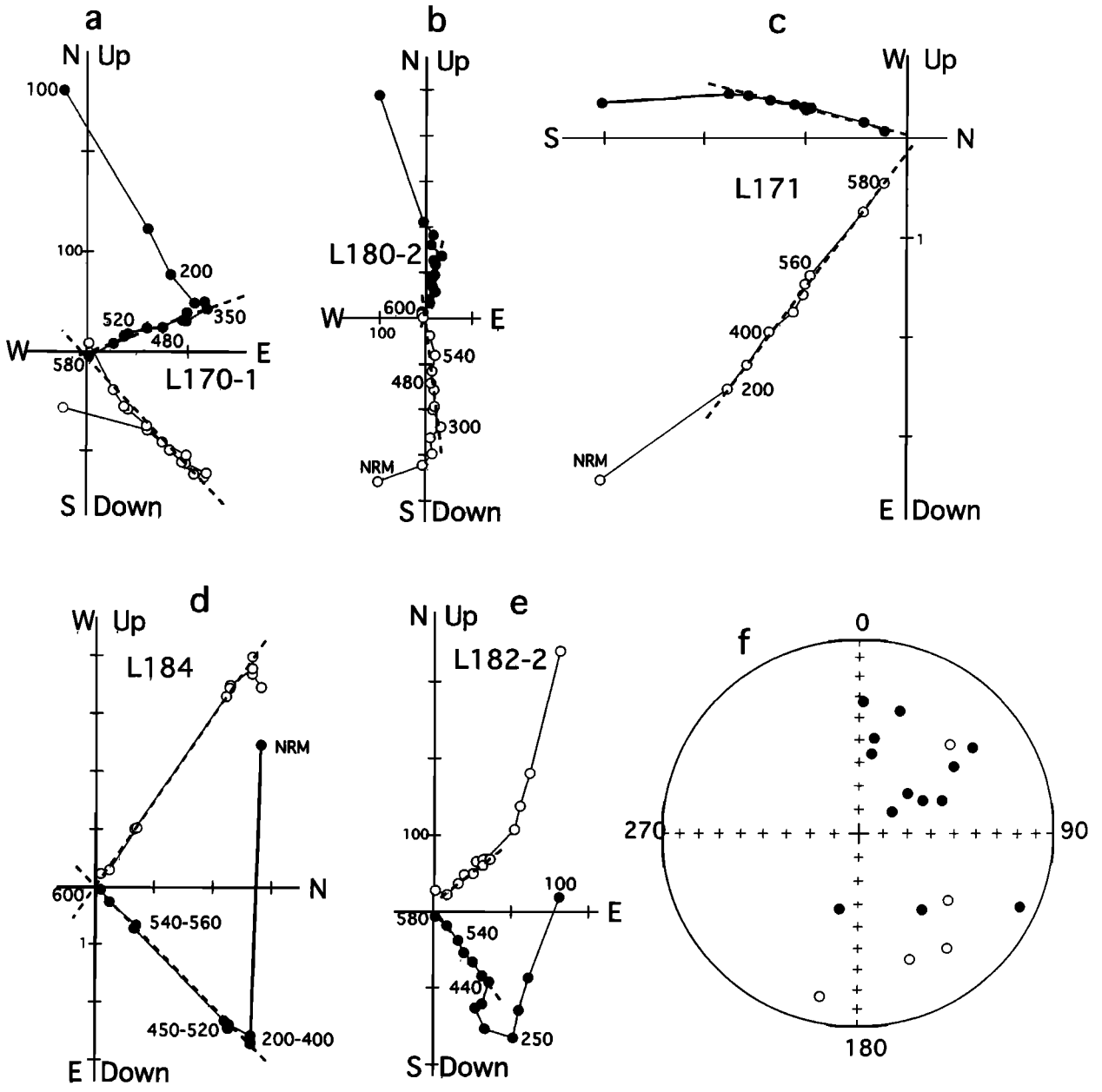


Figure 6. (a-e) Representative vector component plots and (f) stereoplot of ChRM directions for tuffaceous rocks from the lower thrust sheet (site 3). Conventions are as in Figures 3 and 5.

are spot readings of the geomagnetic field, and it is possible to calculate a mean direction for the middle member using just five entries and by treating the result as a site mean. This calculated site mean, however, is poorly defined (Table 1), and its utility is problematic. An important observation is that the thickest stratigraphic interval sampled (site 9) yields a mean direction which agrees well with the results from the lower and upper members, where the stratigraphic intervals sampled vary from several tens of meters to 180 m, and secular variation is better averaged (Table 1). Hence it is not reasonable to pool all the site means with equal weight for calculation of the formation mean; on the other hand, attributing different weights would be arbitrary. It would also be arbitrary to use site 9 and reject the other four simply because of the similarity of the mean vectors. We therefore discarded all data from the middle member.

Comparison of Paleolatitude Data From This Study With Reference Values

The positive fold and conglomerate tests suggest a primary origin of the ChRM in the rocks from the lower and upper members of the nearly 5-km thick Upper Cretaceous-lower Paleocene island arc complex. Relatively thick stratigraphic intervals were studied at the accepted sites and the tight grouping of tilt-corrected site means (Table 1) imply adequate averaging of secular variation. We assume that the overall mean inclination is an unbiased estimate of the geomagnetic field at the place where this complex accumulated.

Tilt-corrected mean inclinations for the lower and upper members of the Khapitsa Formation are statistically indistinguishable, at 65.9° ± 3.8° and 68.4° ± 7.6°, respectively.

These inclinations correspond to paleolatitudes of 48.2° ± 5.0°N and 51.6° ± 10.8°N, respectively. According to the inclination-only analysis, that was applied to the upper and lower members of the Khapitsa Formation. The mean inclination is 66.3° ± 3.7°, which corresponds to a paleolatitude of 48.7° ± 5.0°N.

The apparent polar wander paths for Eurasia and North America [*Besse and Courtillot*, 1991] recalculated to the study area (56°N, 161°E) predict Late Cretaceous paleolatitudes of about 68°-70°N. The observed value is much lower than expected, and northward transportation of the East Range terrane (ERT) is strongly indicated by the data. We examine two end-member models to accommodate such transport: (1) coastwise transport of the island arc terrane after its accretion far to the south of Kamchatka; (2) intraoceanic transport within the Pacific and/or Kula plates. All calculations were made in the hot spot reference frame. Before making kinematic models, however, it is necessary to examine the constraints imposed by geophysical and geological data.

Limits on Tectonic Reconstruction

Limits Imposed by the Plate Motion History

The study area is most likely to have been a part of North America [*Chapman and Solomon*, 1976; *Cook et al.*, 1986; *Riegel et al.*, 1993], and the following discussion is based on this assumption. At 85 Ma the Kula plate formed from the ancestral Farallon plate and was separated from the Pacific plate by a northwest trending transform fault, the location and subsequent movement of which are poorly known; it can only be presumed that the Kula-Pacific transform boundary had been moving westward between 85 and 43 Ma [*Engebretson et al.*, 1985]. Since 66-61 Ma, this boundary was to the east of the East Peninsulas island arc, which probably became active in the central-northern Pacific Ocean [*Pechersky et al.*, 1997].

After 85 Ma the relative movement of the Kula (Pacific) and North American plates was compensated at their convergent boundary; the sense of subduction and the position of the plate boundary, however, are not clear. This configuration probably remained somewhat the same up to 43 Ma, when the Kula-Pacific spreading center became extinct. After 43 Ma the Pacific plate began to move west-northwest, subnormal to the Eurasian margin [*Engebretson et al.*, 1985].

Limits on the Age of the Studied Complex

Paleontological data [*Tsukanov*, 1991] point to a Campanian-Maastrichtian to early Paleocene age for the studied formation, but the ages are not well constrained. As the age of the studied formation and its original paleolatitude derived from paleomagnetic data are crucial for reconstruction of terrane kinematics, the < 20 Myr imprecision in the age of the data renders any consequent analysis unconvincing. It would be all the more true for terranes that could move with the rapidly moving Pacific and/or Kula plates. We tried to further constrain the age of our results using the fact that reversed polarity prevails in the studied rocks. It is unlikely that normal polarity epochs always coincided with hiatuses in sedimentation. Despite the imperfect coverage of the Khapitsa Formation by sampling sites, the dominance of reversed polarity in the collection indicates that the accumulation of this formation most probably correlates with the longest reversed polarity chron in the Late Cretaceous, which are of either early Campanian (83-79 Ma) or mid-Maastrichtian (72-70 Ma) age [*Cande and Kent*, 1992]. It is worth noting that mean inclinations for the lower and upper members of the studied thrust sheet are very similar (Table 1). This observation is also in accord with the above assumption on relatively rapid accumulation of these rocks.

Limits on Docking Time

With major angular unconformity, marine middle Eocene deposits overlie the Upper Cretaceous-lower Paleocene volcanic series, upper Paleogene terrigenous flysch deposits and the Paleocene-lower Eocene Vetlov Formation in the southern part of the ERZ, thus indicating a middle Eocene age for the main deformation event [*Bakhteev*, 1994]. This indicates a middle Eocene upper limit on the docking time of the ERT.

At about 60 Ma the island arc was not far away from the continental shelf, as indicated by the gradual transition from the Khapitsa island arc volcanoclastics to the overlying lower Paleocene continent-derived turbidites. Volcanic glass and pyroxenes systematically decrease through the 15-m transition zone. In contrast, quartz, which was absent in the Khapitsa Formation, appears in increasing concentrations in this zone. Sandstones in the flysch are of varying composition, with high quartz content; their accumulation most likely resulted from erosion of a large continental landmass [*Shapiro et al.*, 1992]. The flysch contains thick lenses of coarse-grained sandstones with angular or semirounded siltstone debris at the base of lenses, which indicate relative proximity to the provenance area. This flysch, though not a proximal deposit sensu stricto, accumulated close to the continental slope, and the island arc was unlikely to have been more than a few hundred kilometers away from the provenance area. Therefore the subduction zone that separated the oceanic plate from a continental plate to which northeast Asia belonged, should have become extinct simultaneously with the cessation of volcanic activity in the related island arc, about 60 Ma. Thus the lower limit on docking of the ERT to the Eurasian margin is constrained by the age of the terrigeneous deposits that overlie the island arc complex.

Cessation of volcanic activity in the ERT island arc nearly coincides with two other events. First, activity on the Lesnov Thrust, which is the suture between the continental margin (West zone) and Upper Cretaceous volcanoclastic complexes (Median Range zone), probably occurred at 65 to 45 Ma. Second, volcanic activity had started at the Kronotsky island arc at the beginning of the Paleocene (66-61 Ma [*Pechersky et al.*, 1997]), thus manifesting the formation of a new subduction zone between northeast Asia and the oceanic plates. Hence, since about 65 Ma, the convergence of continental and oceanic plates was mainly accommodated by subduction under the Kronotsky island arc.

These three events are probably interrelated. The collision of the frontal part of the Cretaceous island arc with the continent probably led to cessation of volcanic activity, active thrusting along this front and accumulation of continent-derived deposits, and the formation of a new subduction zone in its rear because of the need to consume the rapidly moving oceanic plates. If this scenario is correct, it is most likely that at about 65 Ma, the studied terrane was close to the continental margin and that collision had already begun. Thus we can assume that a docking time of 65-60 Ma for the studied terrane is probable.

However, we cannot exclude the possibility that the collision led to the main deformation event in the region. In this

case, collision would have taken place between 55 Ma (the upper limit on the age of the rocks that overlie the island arc complex) and 45 Ma (the lower limit on the age of the rocks that overlie most of the folds and thrusts). Therefore we must consider both docking times. After collision, the ERT probably moved either much more slowly or not at all with respect to northeast Asia.

Coastwise Translation

Paleomagnetic data imply that the studied island arc complex formed at about 20° to the south with respect to its present-day position. In principle, northward drift may result from a displacement within an oceanic plate (or plates) before collision with the continental margin, and/or postcollisional movement along strike-slip faults, as is well established, for instance, for the western margin of North America [*Jarrard*, 1986; *Hagstrum et al.*, 1996]. The main cause of such coast-parallel movement is thought to be oblique collision [e.g., *Kimura*, 1986].

The Kula plate always moved subnormal to the Eurasian margin, provided that the strike of the margin was more or less similar to that at present. Since 43 Ma, the Pacific plate moved to the west northwest, subnormal to the Eurasian margin [*Engebretson et al.*, 1985]. Hence considerable coastwise transport is precluded in these situations (Figure 7). In contrast, convergence of the Pacific plate and northeast Asia was more oblique before 43 Ma, and limited coastwise transport with the Pacific plate was possible. There are no direct geological observations concerning the presence of large sinistral strike-slip faults in Kamchatka; the paucity of outcrops of pre-middle Eocene rocks on Kamchatka, however, does not preclude their existence.

At about 65 Ma a subduction zone, related to the East Peninsulas island arc, became active, and the relative movement of the Pacific and North American plates was mainly

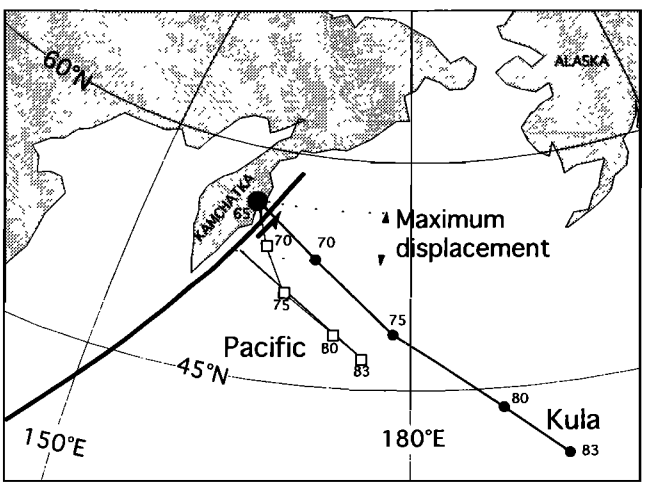


Figure 7. Coastal transport scenarios with the East Ranges terrane being transported with either the Kula (solid dots) or Pacific (open squares) plates from 83-79 Ma until 65 Ma. The double-pointed solid arrow denotes the maximum coastal transport of the East Ranges terrane with the Pacific plate in the direction shown by an solid half arrow. The thick solid line denotes the position of the continental margin. Eurasia and North America are shown in their present-day positions. The trajectories are calculated in the hot spot reference frame using kinematic parameters of *Engebretson et al.* [1985].

compensated at this convergent boundary [*Pechersky et al.*, 1997]; hence any coastwise translation should be older than 65 Ma. The coastwise translation of an island arc while it is active is unlikely, but even if we assume that such movement is possible, the maximum coastal translation of the ERT of less than 4° with the Pacific plate (Figure 7) is much less than the observed difference in latitudes of about 20°. In addition, because the subducting and overriding plates interact in a complex manner, the true coast-parallel velocity should be less than the tangential component of the total velocity [*Beck*, 1986, 1991].

Thus significant coastal transport was impossible after 65 Ma. Even if such transport with the Pacific plate did occur between 83 and 65 Ma, it can account only for a minor part of the total northward displacement of the studied terrane. Coastwise transport also depends on the geometry of the Eurasian margin, which is poorly known. This margin is unlikely to have been as smooth and of such drastically different strike as to allow considerable coastwise transport.

Intraoceanic Transport

There are two possible ages of the studied island arc complex, 83-79 Ma and 72-70 Ma, and two intervals for the docking of the ERT, 65-60 Ma and 55-45 Ma. Because the possibility of considerable coastwise transport was already excluded, we have to assume that the studied terrane was drifting northwestward within a rapidly moving oceanic plate.

Since 65 Ma, the Kula-Pacific transform boundary was to the east of the Kronotsky island arc and related subduction zone which moved with the Pacific plate [*Pechersky et al.*, 1997]. In contrast, between 85 and 65 Ma, the East Ranges island arc could have moved either with the Kula or Pacific plates, because the location of the Kula-Pacific boundary for this period is unclear. It is also possible that the ERT drifted first with the Kula plate and then with the Pacific plate.

As the emplacement could not have taken place before 70 Ma or after 45 Ma, the terrane trajectories can be backward modeled for docking times of 70, 65, 60, 55, 50, and 45 Ma [*Debiche et al.*, 1987]. We calculated trajectories for the both the Eurasian and North American plates and found negligible differences. We consider four scenarios, based on transpositions of the rocks ages and docking times:

Scenario 1: 72-70 Ma Rocks, Docking at 65-60 Ma

Because a terrane of this age could not reach the continental margin at about 65-60 Ma with either the Kula or the Pacific plate (Figure 8a-b), this scenario is rejected.

Scenario 2: 72-70 Ma Rocks, Docking at 55-45 Ma

Volcanic activity started at the Kronotsky island arc at about 65 Ma, thus manifesting formation of a new subduction zone between northeast Asia and oceanic plates. Therefore since that time the convergence of the continental and oceanic plates was mainly accommodated by subduction under this island arc. This reasoning implies that the ERT was moving within either the Kula or the Pacific plate from 72-70 Ma until about 65 Ma; after formation of the Kronotsky island arc and the related subduction zone, it continued moving with some unknown minor oceanic plate, separated from the Pacific plate by the

Kronotsky island arc. There are, however, no traces of this minor plate in Kamchatka. The existence of a minor oceanic plate with the extinct Upper Cretaceous island arc within it necessarily implies the existence of a Paleocene-Eocene subduction zone on the continental margin because the minor oceanic plate had to be consumed somewhere. Paleocene-Eocene volcanism in Kamchatka west of the Kronotsky island arc is scarce and is unlikely be related to a separate subduction zone. This scenario cannot be excluded, but it is improbable.

Scenario 3: 83-79 Ma Rocks, Docking at 65-60 Ma

Movement with the Pacific plate predicts docking times at about 55 Ma (Figure 8a). We conclude that the studied terrane could not have moved permanently with the Pacific plate, and at least a part of its motion was accomplished with the Kula plate. Movement with the Kula plate for a docking time about 65 Ma fits with paleomagnetic data and kinematic parameters within their uncertainties (Figure 8b). A scenario where the studied terrane moved with the Kula plate before ca. 75 Ma and then with the Pacific plate until collision with the continent at about 65-60 Ma (Figure 8c), also fits well with the available data. However, it is not possible to discriminate between the two scenarios. Regardless, both scenarios imply that the Pacific - Kula boundary was to the west of the ERT before 75 Ma and to the east of it after 65 Ma [*Pechersky et al.*, 1997]. Thus this plate boundary had to somehow shift eastward between 75 and 65 Ma.

Scenario 4: 83-79 Ma Rocks, Docking at 55-45 Ma

If the ERT was moving either with the Kula or with the Pacific plate, it had to reach the continental margin long after cessation of volcanism in the East Ranges island arc and the beginning of activity of the Kronotsky island arc. We can presume that when the subduction under the Kronotsky island arc started, the ERT was close to the continental margin but had not yet collided with it. Kinematically, this scenario is indistinguishable from the previous one. It remains, however, unclear why the ERT became anchored without collision and how its final convergence with the Eurasia margin was accomplished. On the basis of the evidence presented above , this scenario is not compelling.

Therefore our preferred interpretation is that the ERT island arc complex formed at 83-79 Ma and arrived at the continental margin at about 65-60 Ma.

Tectonic Models

In addition to the East Ranges island arc, there is also the Campanian - lower Paleocene island arc complex in the Median Range zone of Kamchatka (Figure 1). The abovementioned two complexes are sometimes regarded as parts of one wide island arc [*Zinkevich and Tsukanov*, 1992; *Shapiro*, 1995]. We can also hypothesize the existence of either two coexisting active island arcs or just one island arc, which different parts having docked onto Kamchatka at different times. Accordingly, three models of the tectonic evolution of the northwest Pacific can be suggested for the Late Cretaceous.

The East Ranges island arc was active from 83 -79 Ma until about 60 Ma. Because we assume that the docking time of the ERT was about 65 - 60 Ma, the northward movement of the island arc coincided with its activity. If this movement had been continuous, the related subduction zone also had to move in the absolute reference frame. The absolute motion of a subduction zone should have the same sense of motion as the overriding plate [*Garfunkel et al.*, 1986]. The observed northward movement of the Upper Cretaceous island arc indicates that this arc did not belong to the slowly southward moving North American plate, but rather belonged to a northward moving oceanic plate, most probably the Kula plate. This implies that the related subduction zone was dipping oceanward and was consuming oceanic lithosphere, presumably the oceanic periphery of a continental plate or some minor oceanic plate; a similar geodynamic situation now exists in the New Hebrides trench [*Louat and Pelletier*, 1989]. This process continued until the studied island arc became inactive at about 65 - 55 Ma.

Model 1

We assume that two active island arcs coexisted in the Late Cretaceous. In this case, the Median Ranges island arc was near the continental margin (Figure 9a), and we presume the existence of a minor oceanic plate (which has now completely disappeared) between the two island arcs. We know nothing about the kinematics of this plate and hypothesize that this minor plate had been subducting under the East Ranges island arc on one side and under the Median Range island arc near the continental margin on the other. This process probably stopped by about 65 Ma. At that time, the minor plate should have already been completely subducted.

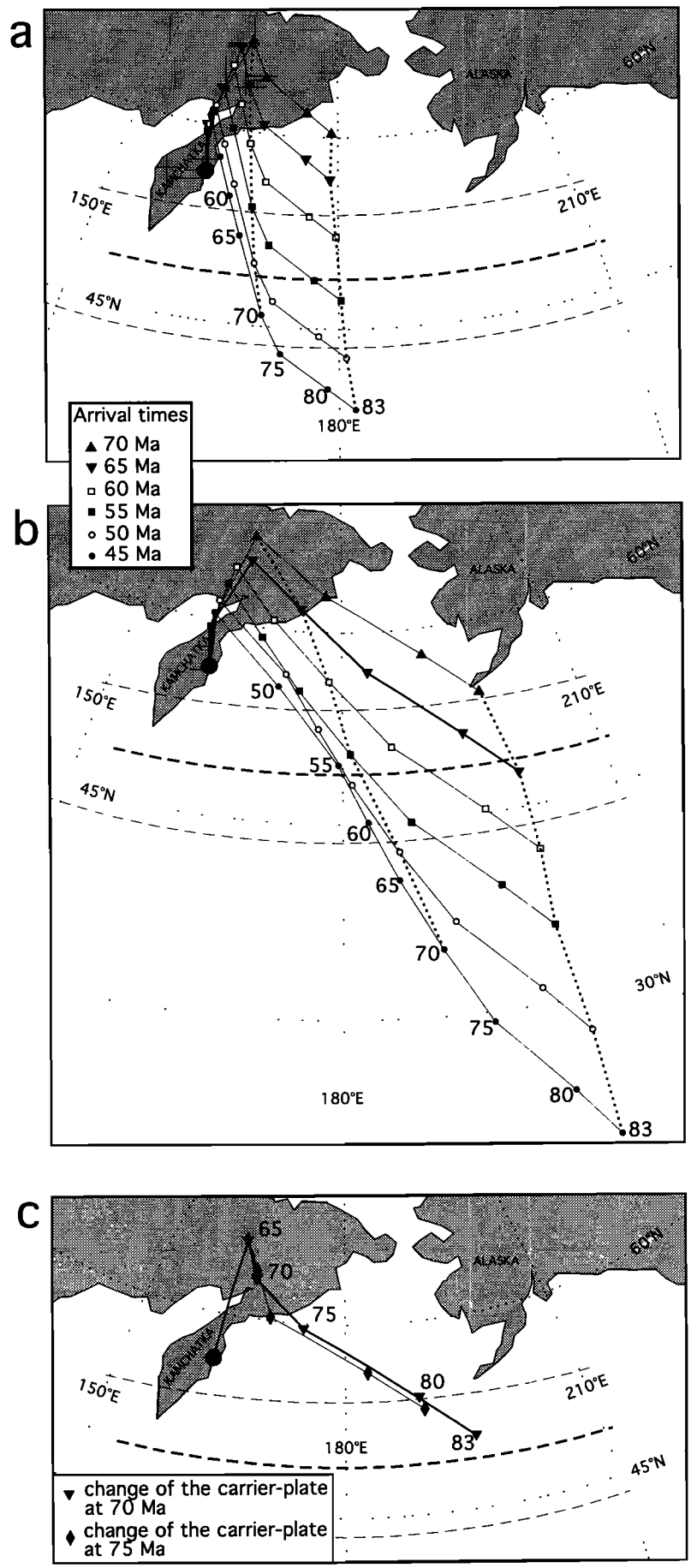
Model 2

In an alternative model the island arc complexes of the Eastern Ranges and the Median Range zones can be treated as part of the same northward moving island arc in the Campanian - Maastrichtian. As discussed above, the Kula - Pacific transform boundary had to be to the west of the ERT before 70-75 Ma and to the east of it after 65 Ma (Figure 9b). This implies that an ancient island arc that comprised the island arc complexes of both the Eastern Ranges and the Median Range zones, was drifting with the Kula plate before 75-70 Ma. At about 75-70 Ma, this island arc was divided into two parts by the Kula-Pacific transform boundary. The western part of this island arc, which is now the Median Range zone, continued moving with the Kula plate until it was emplaced. The eastern part of the island arc (East Ranges zone) then moved with the Pacific plate until its arrival at the continental margin about 65 - 60 Ma (Figure 9b).

Model 3

Finally, the existence of a single wide island arc in the Late Cretaceous also fits the paleomagnetic results and kinematic data. It also accounts for the formation of the Lesnov Thrust simultaneously with cessation of volcanic activity at about 65 Ma. On the other hand, the present-day width of the Median Range and Eastern Ranges zones together is 120 to 150 km, and geological data point to its shortening by factor of 2.5-3 during subsequent folding and thrusting [*Shapiro*, 1995]. The original width of the ancient island arc thus amounts to about 400 km, which is much wider than the modern intraoceanic volcanic zones.

It is difficult to discriminate between the models described above. This situation may become clearer when reliable paleo-


a
ALASKA
60°N
150°E
210°E
45°N
60
65
70
75
80
83
180°E
Arrival times
▲ 70 Ma
▼ 65 Ma
▫ 60 Ma
▪ 55 Ma
◦ 50 Ma
• 45 Ma
b
ALASKA
60°N
150°E
210°E
45°N
30°N
50
55
60
65
70
75
80
83
180°E
c
ALASKA
60°N
150°E
210°E
45°N
180°E
65
70
75
80
83
▼ change of the carrier-plate at 70 Ma
♦ change of the carrier-plate at 75 Ma

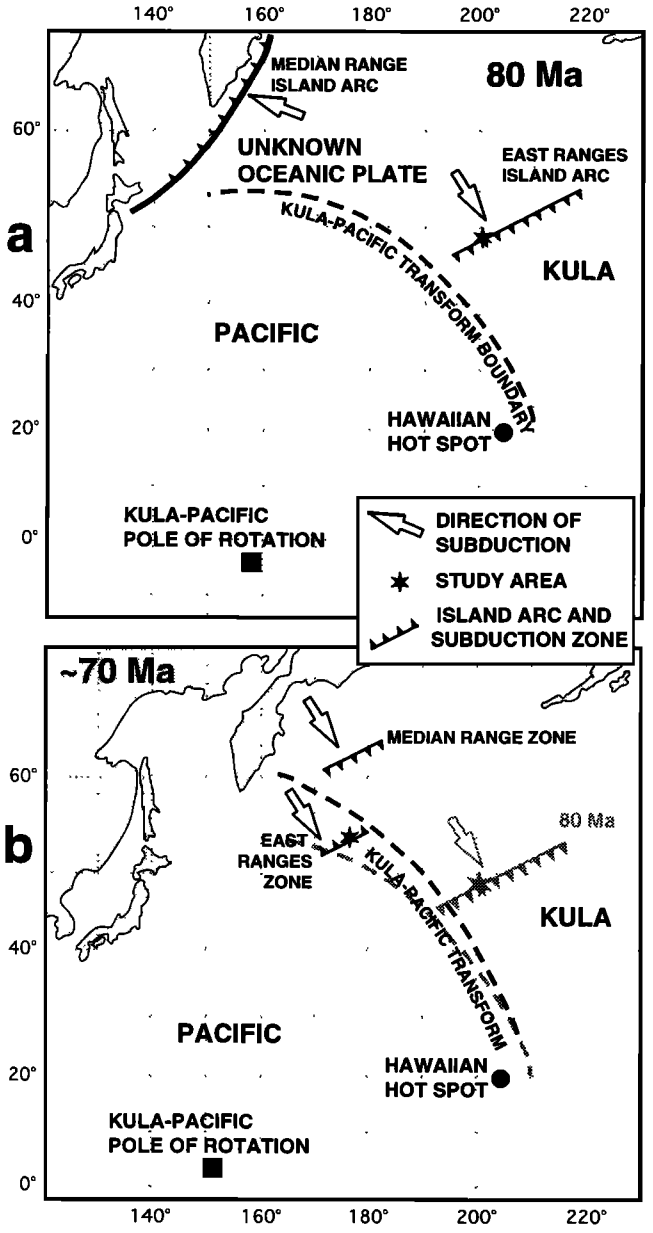


Figure 9. Reconstruction of the northwest Pacific Ocean and its margins at (a) 80 Ma and (b) ca. 70 Ma in the hot spot reference frame using kinematic parameters of *Engebretson et al.* [1985]. In Figure 9a (model 1), two independent subduction zones coexist in the northwest Pacific at 80 Ma. An unknown plate is subducting under the Eurasian margin (Median Range island arc) and the Pacific plate (East Ranges island arc). In Figure 9b (model 2), a single island arc, which had existed at an observed paleolatitude at 80 Ma, is cut into two segments that are carried by different plates by the eastward migrating Kula-Pacific transform boundary. These segments docked onto the Eurasian margin at different times, which accounts for the existence of two island arc complexes (Median Range and East Ranges zones). The 80 Ma positions of the East Ranges island arc and Kula-Pacific transform boundary are shown in light gray.

magnetic data from well-dated rocks of the Median Range island arc complex are available.

Conclusions

The prefolding, probably primary, characteristic remanence that was isolated from the Late Cretaceous island arc complex of the Eastern Ranges zone of Kamchatka suggests that these rocks formed at a latitude of 48.7° ± 5.0°N which is about 20° south of the latitudes predicted for the Eurasian margin. Geological and paleomagnetic data imply that the most likely age of the studied rocks is 83-79 Ma. The Eastern Ranges island arc probably collided with, or was close to, the Eurasian margin at about 60-65 Ma. Kinematic calculations predict that this terrane was in the middle of the central-northern Pacific Ocean at 83-79 Ma, and from that time until 65-60 Ma, it was moving northwestward either with the Kula plate or with the Kula and Pacific plates consecutively. Since approaching the continental margin at 65-60 Ma, there were no significant movements of this terrane until the time of complete emplacement.

The suggested tectonic scenario is in accord with geological data. Volcanic activity ceased at about 65-55 Ma. The continent-derived Paleocene turbidites, which accumulated close to the continental slope, overlie the island arc complex with gradual transition, thus implying that the island arc became extinct when it was close to the continent. The formation of the Lesnov Thrust, the suture between the continental margin and the Upper Cretaceous arc-related complex, most probably occurred at 65-55 Ma. Because the convergence between the continental and oceanic plates was mainly accommodated by subduction under the Kronotsky island arc since 65 Ma, there is no likely mechanism for significant drift of the studied island arc after 65 Ma. Therefore, we assume that thrusting in central Kamchatka probably resulted from accretion of the Campanian-lower Paleocene island arc.

Eocene deformation (ca 45 Ma) affected the Kumroch Range as well as most of Kamchatka. This deformation may have resulted from a jump of the subduction zone from its position close to the Kronotsky arc in the central-northern Pacific Ocean [*Pechersky et al.*, 1997] to a new position along the Eurasian continental margin behind the already accreted Upper Cretaceous-lower Paleocene arc(s). This new subduction zone, which was responsible for Oligocene-Miocene volcanism in central Kamchatka, was active until the late Miocene.

In conclusion, it is evident that paleomagnetic data enable the tectonic evolution of Kamchatka to be deciphered and that they also help to constrain the kinematics of oceanic plates, as is demonstrated for the Kula-Pacific transform boundary. Further interpretation is limited not only by the scarcity of paleomagnetic data but by poor age constraints on the main geological events that formed Kamchatka. Our understanding of the accretionary tectonics of northeast Asia and the kinematics of oceanic plates will greatly depend on obtaining reliable new paleomagnetic data and extensive work that must be directed at better dating of geological events.

Figure 8. Motion of the East Ranges island arc with (a) the Pacific plate and (b) the Kula plate for different arrival times between 70 and 45 Ma, and with (c) first the Kula plate and then the Pacific plate for an arrival time of 65 Ma. The trajectories (solid lines) are backward modeled [*Debiche et al.*, 1987] using the kinematic parameters of *Engebretson et al.* [1985]. The observed paleolatitude and its confidence limits are shown as thick and thin dashed lines, respectively. Thick dotted lines denote isochrons in Figures 8a and 8b. The trajectories are calculated in the hot spot reference frame. Kamchatka is assumed to belong to the North American plate. Eurasia and North America are shown in their present-day positions.

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